TECHNICAL SPECIFICATION FOR CABLES

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I. 25 Sq mm Battery Cables

1.0 SCOPE:

1.1. The scope of the specification covers design, manufacture, testing, packing, forwarding and delivery on FADS basis of cables as detailed below

i) 25 Sq.mm. Battery cable.

1.2. The battery cable is intended to interconnect the (-)48V DC battery to battery charger carrying currents of about 100A. Battery cables are intended for indoor use only.

2.0 **STANDARDS:**

(i) The cables proposed for purchase shall conform to the following I.S. Standards which shall mean latest revisions, amendments/changes adopted and published unless otherwise specified herein after.

Standards	Title
	Battery cable
IS1554(Part1) 1988 Category	PVC insulated (Heavy duty) Electric cable Part-1, for working voltages up to and including 1100V.
IS 8130/1984	Conductors for insulated electric cables and flexible cables.
IS5831/1984	PVC insulation and sheath of electric cable.

Equipment meeting with the requirement of other authoritative standards, including IS which ensure equal or better performance than the standards mentioned above, shall also be considered. When the equipment offered by the bidder conforms to other standards, salient points of difference between standards adopted and the standards specified in this specification shall be clearly brought out in the relevant schedule. Four copies of such standards with authentic translation in English shall be furnished along with the offer.

3.0 CLIMATIC CONDITION :

3.1 The Cables called in this specification are required to operate satisfactorily as per the following service conditions.

i.	Location	:	Andhra Pradesh
ii.	Max ambient air temperature(deg.C)	:	55
iii.	Minimum ambient temperature(deg C)	:	7.5
iv.	Average daily ambient air temp(deg.C)	:	35
v.	Maximum relative humidity(%)	:	74
vi.	Average annual rainfall(mm)	:	925
vii.	Maximum wind measure(kg/m2)	:	200
viii.	Max attitude above mean see level(m)	:	1000
ix.	Ceraunic level (days/year)	:	40 to 50
x.	Seismic level (horizontal acceleration)	:	0.10g

4.0. PRINCIPAL PARAMETERS:

4.1.0. 25 Sq. mm single core multi strand battery cable

4.1.1.	Type of cable	: 25 Sq.mm FRLS type Category C1 conforming to IS 1554(Part1)battery cable
4.1.2.	Conductor material	: Multistrand annealed high conductivity bare copper as per IS:8130/84
4.1.3.	No. of strands in the conductor /dia of each strand	: 209/0.4 mm
4.1.4.	Insulation material	: PVC Type-C FRLSH
4.1.5.	Thickness of Insulation	: 1.2mm Nominal
4.1.6.	Approx. cable outer dia	: 11.0 mm Nominal
4.1.7.	Max. CR at 20 deg.C.	: 0.95 Ohms/km
4.1.8.	Volume resistivity	: 1 X 10^{12} ohms/cms at 20 deg.C
4.1.9.	High voltage test	: 3.0 kV for 5 minutes between conductor and ground
4.1.10	RLS type	: Category C1
4.1.11	Oxygen Index	: As per ASTMD 2963
4.1.12	Temperature Index	: not less than 25^0C

5.0. GENERAL TECHNICAL REQUIREMENTS

5.1. The Battery cables shall be armored. Battery cable conductor shall be multi strand annealed high conductivity bare copper as per IS:8130/84.The overall FR of category C1 conforming to IS-1554(Part1) and PVC insulation type A of IS:5831/1984 may generally conform.

5.2. The cable shall be designed and manufactured so that the damage will not result from transportation installation and operation under any/or all the climatic and operating conditions to which they may be subjected to.

5.3. Cable shall be suitable for laying in conduits, ducts, trenches and trays.

5.4. The cable shall be supplied in the minimum length of 500m with tolerance of +/- 5% per coil to a total overall tolerance of +/- 5% of the total ordered quantity.

5.5. Half the quantity may be supplied in Red colour PVC outer sheath and the other half with Black colour PVC sheath.

6.0. TESTS

6.1. The type tests carried out on the offered cables should be in accordance with IS:5026-1969, IS:11967(part.2/sec.2)/89,

6.1.1. The bidder along with his bid shall furnish type test certificates for the tests specified in clause No.6.2.0 below for the equipment offered by him. The type tests should have been conducted on particular type & model of the equipment i.e., offered by the bidder against this specification by a reputed independent laboratory and the satisfactory performance of the said equipment should have been certified by the laboratory.

The bids of bidders not accompanied by the type test certificates as stated above will be treated as incomplete and termed as non-responsive and liable to be rejected.

Type tests shall mean those tests, which are to be carried out to prove the process of manufacture and general conformity of the material to this specification.

6.2. Type Tests

i) dimensions

ii) elongation test

iii) tensile test

iv) insulation resistance test

v) high voltage test

vi) flammability test

vii) spark test

viii) resistivity of the conductor

ix) Oxygen index and temperature rise test as per ASTMD-2863.

The minimum value of oxygen index shall be 29.

x) Flame retardant test on single cable.

xi) Rodent and termite proof test.

6.3. Acceptance Tests

i) dimensions

ii) insulation resistance

iii) elongation test

iv) tensile test

v) high voltage test

vi) conductivity test / shrinkage

vii) thermal stability test

6.4 Routine Tests

i) dimensions

ii) insulation resistance

iii) elongation test

iv) tensile test

v) high voltage test

vi) conductivity test/ shrinkage

6.5. Testing Expenses

The entire cost of testing for the acceptance and routine tests and tests during manufacture specified herein shall be treated as included in the quoted unit price of cables.

6.6. Additional Tests

The Purchaser reserves the right of having at his own expenses any other test(s) of reasonable nature carried out at Bidder's premises, at site, or in any other place in addition to the aforesaid type, acceptance and routine tests, to satisfy himself that the material comply with the specifications.

6.7. Test Reports

6.7.1. Copies of test reports shall be furnished in at least six (6) copies along with one original. One copy shall be returned duly certified by the Purchaser only after which the material will be dispatched.

6.7.2. Record of routine test reports shall be maintained by the Bidder at his works for periodic inspection or as and when desired by the Purchaser's representative.

6.7.3. Test Certificates of tests conducted during manufacture shall be maintained by the Bidder. They shall be produced for verification as and when desired by the Purchaser.

6.8. TEST FACILITIES

6.8.1. The following additional test facilities shall be available at Bidder's works:

a) Various testing and measuring equipment shall be provided.
b) Standard resistance for calibration of resistance bridges.

GUARANTEED TECHNICAL PARTICULARS
for Battery Cable
(to be filled by the bidder)

	TESTS		**25 Sq.mm**
1.	Manufacturers Name & Address	;	
2.	Suppliers' Name & Address	:	
3.	Governing Standards	:	
4.	Type of Cable	:	
5.	Category of Cable	:	
a.	Oxygen Index (Reference Standard)	:	
b.	Temperature Index(Reference Standard)	:	
6	No. of core	:	
7.	Voltage rating	:	
8.	Conductor/Strands		
	a) material	:	
	b) diameter	:	
	c) No. of strands	:	
9.	Insulation		
	a) material	:	
	b) thickness	:	
	c) tolerance on thickness	:	
	d) colour of core	:	
10.	Approx. cable outer dia.	:	
11.	Construction details	:	
12.	Other constructional details	:	
13.	Min. installation radius		:
14.	Electrical test		
	a) Max.CR at 20 deg.C.	:	
	b) Volume resistivity at 27 deg.C .	:	
	c) High voltage	:	
15.	Packing length	:	
16.	Whether type test certificates enclosed:		Yes/No

Date :
Place :

Signature of Manufacturer/Supplier
Name & Address

Signature of the Bidder
Name :
Whether Authorised Attorney of the tendering company
Name of the tendering Company with Seal

2.5 Sq.mm UN-ARMOURED COPPER CONTROL CABLE

1.0 SCOPE :

1.1 This Specification covers design, engineering, manufacture, stage testing, inspection and testing before dispatch, packing & forwarding and delivery FADS destination including unloading, of 2.5 sq.mm un-armored multi core, PVC insulated and sheathed copper control cables for use on AC single phase or three phase (earthed or unearthed) systems of rated voltages up to and including 1100 V or DC systems of rated voltages up to and including 1500 V to earth specified herein for their satisfactory operation in different substations in Andhra Pradesh in India.

Sizes of the LT copper control cable are

i) 2c x 2.5 sq.mm

ii) 4c x 2.5 sq.mm

iii) 6c x 2.5 sq.mm

iv) 10c x 2.5 sq.mm

v) 12c x 2.5 sq.mm

The cables shall conform in all respects to high standards of engineering, design and workmanship and latest revision of relevant standards at the time of offer. The cables shall be capable of performing in continuous commercial operation up to the Bidder's guarantee acceptable to the Purchaser, who will interpret in a manner the meanings of drawings and Specifications and shall have the power to reject any work or material which in his judgment is not in accordance therewith.

Sl No	Standard	Title
1.	IS: 1554	Specification for PVC insulated (heavy duty) electric cables for working voltages up to and including 1100 Volts
2.	IEC: 228	Conductor for insulated cables
3.	IS:8130	Specification for conductors for insulated electric cables and flexible cords
4.	IS:5831	Specification for PVC insulation and sheath of electric cables

5	IS:10810	Specification for methods of test for Cables
6.	IS:3961 (Part-V)	Recommended current ratings for cables (PVC insulated light duty cables)
7.	IS:3975	Specification for Galvanized Steel wires, Formed wires and Tapes for armoring of cables
8.	IS:10418	Specification for Drums for Electric Cables
9	IS:613	Specification for Copper Rods and Bars for Electrical Purposes

1.2 STANDARDS :

Unless otherwise specified elsewhere in this specification, the control cables shall conform to the latest revisions with amendments with available of relevant standards, rules and codes some of which are listed hereunder for ready reference, unless otherwise specified elsewhere in this specification.

REFERENCE STANDARDS	NAME AND ADDRESS
IS	(Indian Standard) Bureau of India Standards, Manak Bhavan 9, Bahadurshah Zafar Marg, New Delhi - 110 002, INDIA
IEC	(International Electro Technical Commission) Bureau Central de la Commission, Electro Technique International 1 Rue de verembe, Grneva, Switzerland
ISO	(International Organisations for Standardization) Danish Board of Standardization Danish Standardizing Sraat Aurehoegvaj - 12, DK-2900 Heelprup DENMARK

Cables meeting with the requirements of other authoritative National or International Standards, which ensure equal or better performance than the standards mentioned above, shall also be considered. When the equipment offered by the bidder conforms to other standards salient points of difference between standards adopted and the standards specified in this specification shall be clearly brought out in the relevant schedule. Four copies of such standards with authentic translation in English shall be furnished along with the offer.

2.0 SERVICE CONDITIONS :

2.1 CLIMATIC CONDITIONS:

Equipment/ material to be supplied against this specification shall be suitable for satisfactory continuous operation under the climatic / tropical conditions specified in clause-23.1 of section-III of this specification.

3.0 MAIN FEATURES :

3.1 These General Purpose Insulation cables shall be suitable for use where the combination of ambient temperature and temperature rise due to load results in conductor temperature not exceeding.

- 70°C for normal continuous operation
- 160°C for short circuit condition

The cables shall be suitable for laying in covered/open trenches, ducts, trays or buried directly underground under dry or wet conditions. Suitable additives to prevent attack by rodents, termites shall be added to the outer sheath PVC compound. Also, the outer sheathing shall be so designed as to afford high degree of mechanical protection besides making it resistant to heat, oils, chemicals, abrasion, weather conditions etc. Common acids, alkalis, contaminants saline solutions etc., shall not have adverse effects on the PVC sheathing material used.

3.2 The cables shall be capable of operating continuously under the system frequency variation of ±5% voltage variation of ±10% and a combined frequency - voltage variation of ±10%.

4.0 TECHNICAL REQUIREMENTS :

4.1. CONDUCTOR

4.1.1 MATERIAL:

The conductor shall be composed of plain annealed copper wires (complying with IS:8130) made of high conductivity copper rods (complying with the latest version of IS:613).

4.1.2 FORM:

The conductors shall be circular, stranded or sector shaped. They shall be clean, reasonably uniform in size and shape, smooth and free from all harmful defects.

4.1.3 JOINTS:

Joints shall be permitted in the individual wires of which the conductor is formed but no joint shall be within 300 mm of any other joint within the same layer. The joints shall be made by resistance butt welding, fusion welding, cold pressure welding, electric welding, gas welding, brazing or silver soldering.

4.1.4 Classification:

Class-2 for stranded circular non-compacted cables used for fixed installation.

4.1.5 SIZE, NUMBER, RESISTANCE:

The wires in the conductor shall have the same nominal diameter before stranding. The number of wires in the conductor shall be not less than 7 (seven). The minimum diameter of each wire after stranding at any point, through-out the length of the cable shall not be less than the nominal diameter by 0.01 mm. The maximum resistance of conductor at 20 deg. C shall be 7.41 Ohms/km as per Table 2 of IS: 8130.

4.2 INSULATION :

4.2.1 MATERIAL:

The insulation shall be "General Purpose Insulation" of Type-C PVC Compound (conforming to the requirements of IS-5831) suitable for a maximum rated conductor temperature of 70°C intended for cables with rated voltages up to and including 3.3kV. The conductor shall be provided with this Type-C PVC insulation applied by extrusion.

4.2.2 COMPOSITION:

The insulation shall consist of a compound based on one of the following materials, which have been SUITABLY COMPOUNDED AND PROCESSED to meet the requirement of the IS:5831.

a) Polyvinyl chloride (PVC)

b) Suitable co-polymers of which the major constituent shall be vinyl chloride.

c) Mixture of (a) and (b)

4.2.3 Test Requirement: The insulation shall satisfy the test requirements stated in columns 2,3,4, & 7 of TABLE-1 of IS:5831.

4.2.4 THICKNESS: The average thickness of insulation shall be not less than the nominal value (t1) of 0.9 mm applicable for unarmoured multi-core cables of size 2.5 sq.mm as per column 3 of Table 2 as specified in IS:1554 (Part I).

4.2.5. Tolerance on thickness: The smallest of the measured values of thickness of insulation shall not fall below the nominal value (t1 in. mm) by more than 0.1 mm + 0.1 t1.

4.2.6 Application of Insulation: The insulation shall be so applied that it fits closely on the conductor and it shall be possible to remove it without damage to the conductor.

4.3 CORE:

4.3.1 Core Identification **:** Cores shall be identified by different colouring of PVC insulation by adopting the following scheme:

For 2 cores: Red and Black

For 4 Cores: Red, Yellow, Blue & Black.

For 6/10/12/7/14/19/27 cores: As per standards. Two adjacent cores (counting and direction core) in each layer, blue and yellow remaining cores gray. By numbers in which case the insulation of all the cores shall be of same colour and numbered sequentially, starting with number-1 for the inner layer. All the numbers shall be of the same colour which shall contrast with the colour of the insulation. The numerals shall be legible.

4.3.2 Arrangement of Marking: The numbers shall be repeated at regular intervals along the core, consecutive numbers being inverted in relation to each other. When the number is a single numeral, a dash shall be placed underneath it. If the number consists of two numerals, these shall be disposed one below the other and a dash placed below the lower numeral. The spacing "d" between consecutive numbers shall not exceed 50 mm as shown in fig.1 of IS: 1554 (Part-I)-1998 Page-4 (core identification by numbers).

4.3.3. LAYING UP OF CORES: In twin and multi-core cables, the cores shall be laid up together with a suitable lay; the outermost layer shall have right hand lay and the successive layers shall be laid with opposite lay, where necessary the interstices shall be filled with non-hygroscopic material.The lay-up of core shall be applicable as recommended in Table-3 of the IS: 1554 Part-1.

4.3.4 INNER SHEATH:

a) MATERIAL: Filler and Inner sheath shall be as follows:

a) Unvulcanised rubber or

b) Thermoplastic materials, or

c) Proofed tape (for inner sheath only)

Unvulcanised rubber or thermoplastic material used shall not be harder than the type-C PVC used for insulation and the Type ST-1 PVC used for outer sheath. The material shall be so chosen as to be compatible with temperature ratings of the cable and shall have no deleterious effect on any other component of the cable.

a) CONSTRUCTION: The laid up cores shall be provided with inner sheath applied by extrusion only. It shall be ensured that the shape is as circular as possible. The inner sheath shall be so applied that it fits closely on the laid up cores and it shall be possible to remove it without damage to the insulation. The thickness of the inner sheath shall be as specified in column-3 of Table-4 of IS:1554, applicable for the multi core cables whose

calculated diameter over laid up cores is up to and including 25 mm and above. If and when one or more layers of binder tapes are applied over the laid up cores, the thickness of such tapes shall not be construed as part of the inner sheath.

4.4. OUTER SHEATH:

4.4.1 MATERIALS: For cables with general purpose insulation, the outer sheath shall be of Type ST-1 PVC compound conforming to the requirements of IS:5831 as recommended for general purpose sheath intended for use in cables operating at a maximum rated conductor temperature 70 deg C. The test requirements shall be as per Columns 2, 3, 4 & 6 of Table-2 of IS: 5831.

4.4.2 CONSTRUCTION: The outer sheath shall be applied by extrusion over the inner sheath as applicable for un-armoured twin and multi core cables. It shall be so applied that it fits closely over inner sheath and it shall also be possible to remove it without damaging the inner sheath. The colour of the outer sheath shall be black.

4.4.3 THICKNESS: The average and minimum thickness of the PVC outer sheath shall be not less than the values specified in col.3 & 4 of Table 7 of IS: 1554.

5.0 TESTS:

5.1. TYPE TESTS: The cables offered should have been fully type tested as per IS/IEC (with latest amendments) in a recognized laboratory. The date of type tests conducted shall be the latest and not earlier than Ten years as on the date of bid opening. The bidder shall furnish type test reports along with the offer. The bids received without type test reports will be considered as **non responsive.**

5.2 ACCEPTANCE TESTS: The following shall constitute acceptance tests.

a) Annealing test,

b) Conductor resistance,

c) Thickness of insulation and sheath,

d) Tensile strength and elongation at break of insulation and sheath,

e) Insulation Resistance Test,

f) High Voltage Test at room temperature,

The sampling plan for acceptance tests shall be as per Appendix-A of the IS:1554 (Part-1)

5.3 ROUTINE TESTS: The following shall constitute routine tests:

a) Conductor resistance test

b) High voltage test at room temperature

5.4 OPTIONAL TESTS: a) Cold bend and b) Cold impact

NOTE: i) For test methods, relevant parts of IS: 10810 shall be referred,

ii) The test certificates as a result of the inspection of the cables shall be furnished to the purchaser for his approval prior to despatch.

6.0. INSPECTION

6.1 The Purchaser's representative shall, at all times, be entitled to have access to the works and all places of manufacture where cable shall be manufactured and the representative shall have full facilities for unrestricted inspection of the Bidder's works, raw materials and process of manufacture and conducting necessary tests as detailed herein.

6.2 The Bidder shall keep the Purchaser informed in advance of the time of starting and of the progress of manufacture of cable in its various stages so that arrangements can be made for inspection.

6.3 No material shall be despatched from its point of manufacture before it has been satisfactorily inspected and tested, or unless the inspection is waived by the Purchaser in writing. In the later case also, the cable shall be despatched only after satisfactory testing for all tests specified herein has been completed and approved by the Purchaser.

6.4 The acceptance of any quantity of material shall in no way relieve the Bidder of any of his responsibilities for meeting all requirements of the specification, and shall not prevent subsequent rejection if such material is later found to be defective.

6.5 At least 5% of the total number of drums subject to minimum of two in any lot put up for inspection shall be selected at random to ascertain the length of cable by the following method:

"At the works of the manufacturer of the cable, the cable shall be transferred from one drum to another at the same time measuring its length with the help of a graduated pulley & Cyclometer. The difference in the average length thus obtained and as declared by the Bidder in the packing list shall be applied to all the drums if the cable is found short during checking."

6.6 The bidder shall give 15 days advance intimation to enable the purchaser to depute his representative for witnessing acceptance and routine tests. All the charges in connection with inspection by purchaser personnel such as travel, accommodation and incidentals shall be borne by the purchaser.

7.0 QUALITY ASSURANCE PLAN :

7.1 The bidder shall invariably furnish the following information along with his offer failing which the offer shall be liable for rejection. Information shall be separately given for individual type of equipment offered.

a) The Structure of organization

b) The duties and responsibilities assigned to staff ensuring quality of work

c) The system of purchasing, taking delivery and verification of materials

d) The system for ensuring quality of workmanship

e) The quality assurance arrangements shall confirm to the relevant requirement of ISO-9001 on ISO 9002 as applicable.

f) Statement giving list of important raw materials, names of sub-supplies for the raw materials, list of standards according to which the raw material are tested, list of tests normally carried out on raw material in the presence of suppliers representative, copies of test certificates.

g) Information and copies of test certificates in respect of bought out items.

h) List of manufacturing facilities available

i) Level of automation achieved and list of areas where manual processing exists.

j) List of areas in manufacturing process, where stage inspections are normally carried out for quality control and details of such tests and inspections.

k) The List of testing equipment available with the bidder for final testing of equipment specified and test plant limitation, if any vis-a-vis the type. Special acceptance and routine tests specified in the relevant standards. These limitations shall be very clearly brought out in "Schedule of Deviations" from the specified test requirements.

7.2 The Contractor shall within 30 days of placement of order, submit the following information to the purchaser.

a) List of the raw material as well as bought out accessories and the names of sub-suppliers selected from those furnished along with the offer.

b) Type test certificates of the raw material and bought out accessories if required by the purchaser.

c) Quality Assurance Plan (QAP) with hold points for purchaser's inspection. QAP and purchasers hold points shall be discussed between the purchaser and contractor before the QAP is finalized.

The contractor shall submit the routine test certificates of bought out accessories and central excise asses for raw material at the time of routine testing if required by the purchaser and ensure that the quality assurance requirements of specification are followed by the sub-contractor.

7.3 The Quality Assurance Programme shall give a description of the Quality System and Quality Plans with the following details.

a) Quality System:

i) The Structure of the Organisation

ii) The duties and responsibilities assigned to staff ensuring quality of work.

iii) The system of purchasing, taking delivery of verification of materials.

iv) The system of ensuring of quality workmanship.

v) The system of control of documentation.

vi) The system of retention of records.

vii) The arrangement of contractor internal auditing.

viii) A list of Administrator and work procedures required to achieve contractors quality requirements. These procedures shall be made readily available to the purchaser for inspection on request.

b) Quality Plans:

i) An outline of the proposed work and program sequence.

ii) The structure of contractors organizations for the contract.

iii) The duties and responsibilities ensuring quality of work

iv) Hold and notification points.

v) Submission of engineering documents required by this specification.

vi) The Inspection of the materials and components on request.

vii) Reference to contractors work procedures appropriate to each activity.

viii) Inspection during fabrication/construction.

ix) Final inspection and test.

8.0 DOCUMENTATION:

8.1 The sectional view of control cable drawings to be offered against the bid are enclosed vide annexure-II.

In addition to any other drawings which the bidder may like to upload, the following manuals must be submitted along with the tender.

a) Drawing showing sectional view of cables and cable drums

b) Technical and descriptive literature giving details of the cables offered

c) Tables of rating factor for correction of current rating for variation in standard condition of installation.

d) Test facilities available at the bidder's works.

8.2 TEST REPORTS:

a) Four copies of acceptance test reports shall be furnished to the purchaser. One copy will be returned, duly certified by the purchaser and only thereafter shall the materials be despatched.

b) All records of routine test reports shall be maintained by the Bidder at his works for periodic inspection by the purchaser.

c) All test reports of tests conducted during manufacture shall be maintained by the Bidder. These shall be produced for verification as and when requested for by the purchaser.

9.0 IDENTIFICATION, PACKING AND MARKING:

9.1.1 MANUFACTURER'S IDENTIFICATION:

The manufacturer shall be identified throughout the length of the cable by means of a tape bearing the manufacturer's name or trade mark, by being indented, printed or embossed on the outer sheath of the cable.

9.1.2 CABLE IDENTIFICATION:

The words "APTRANSCO" shall be embossed and printing of other details shall be done with Inkjet/laser printer throughout the length of the cable at frequent intervals (not more than 1 meter in any case) on the outer sheath.

9.1.3. Cumulative Length of the Cable in metres shall be indented, printed or embossed throughout the length of the cable starting from '0' at each interval of one metre length.

9.1.4 **CABLE CODE:**

Cable code shall be as per IS: 1554 to indicate Copper cable having PVC insulation PVC outer sheath.

9.2.1 PACKING:

The cable shall be wound on a non-returnable wooden/steel drum (conforming to the latest version of IS: 10810) of suitable size and packed. The ends of the cable shall be sealed by means of non-hygroscopic sealing material. The drum packing shall be robust enough for rough handling that may be subjected to during transit by water/rail /road. Type and material of the drum shall be specified by the Bidder by means of a dimensional drawing.

9.2.2 STANDARD DRUM LENGTHS AND QUANTITY TOLERANCE:

Each cable drum shall contain one single continuous length. The desired drum lengths shall be 1000 metres ± 10 % for 2/ 4/ 6/7 Core cables and 1000/500 metres ± 10 % for 10/12/ 14/19/27core cables. The overall quantity tolerance shall be ± 5 %.

The Purchaser reserves the right to alter the drum lengths, if required, while placing the order. The bidder may also offer the maximum drum lengths he is capable of supplying against each size.

9.3 DRUM MARKING:

"THE CABLE DRUM SHALL BE DULY "IS" MARKED" and carry the following details:

a) Reference to the "IS"

b) Manufacturer's name, brand name, trade name

c) Type of cable and voltage grade

d) Number of cores

e) Nominal cross-sectional area of the conductor (Sq.mm).

f) Cable code.

g) Color of cores

h) Length of the cable on the drum (Metres)

i) Direction of rotation of drum by means of arrow.

j) Approximate gross mass (Kg)

k) Running end of cable

l) Country of manufacturer

m) Year of manufacture

n) Standard Mark

10.0. GUARANTEED TECHNICAL PARTICULARS

The Guaranteed Technical particulars (GTP) of the material offered shall be as per Annexure-1 of this specification and the bidders shall furnish the duly signed copy of this GTP. The bids received without the duly signed GTP (Annexure-I) shall be treated as non responsive and rejected.

III. CAT6 UTP CABLE

S.No	Cat 6 UTP Copper cable	
1	CAT6 UTP cable	Cat.6, U/UTP cable. 100 Ohm impedance. Data transmission frequencies of upto 250 MHz.
2	Standardization	ISO/IEC 11801 ed. 2.2; IEC 61156-5 2nd ed.;EN 50173-1; EN 50288-6-1; TIA 568-B.2.1; PVC.
3	Application	Primary (Campus), Secondary (Riser), Tertiary (Horizontal) IEEE 802.3: 10Base-T; 100Base-T; 1000Base-T; IEEE 802.5 16 MB; ISDN; TPDDI; ATM
4	Category	CAT6
5	Conductor	bare copper wire Ø 0.57 mm
6	No of conductor	8
7	Insulation	Polyethylene, Ø 0.90 mm
8	Twisting	2 cores to the pair
9	Cable lay up	4 pairs to the core
10	Cu cable shielding	U/UTP
11	Stranding	4P
12	Cable jacket material	PVC
13	Cable jacket characteristics	cable, metal-free ; flame-retardant
14	Cable overall diameter	Ø 5.6 mm ± 0.4
15	Conductor diameter	AWG23
16	Conductor Type	Tight buffer
17	Length (meter)	305
18	Tensile force	100N
19	Mechanical properties Bending radius	>=35 mm without load, >= 55 mm with load.
20	Temperature range	during operation -20°C+ 60°C, during installation 0°C+ 50°C.
21	Electrical properties DC loop resistance	(at 20°C ± 5°C): =< 14.5 W/100m.
22	Resistance unbalance	=<2%
23	Insulation resistance (500 V)	>= 5000 M ohm km.
24	Capacitance	44 pF/m max
25	Capacitance unbalance (pair to ground)	=<500 pF/km max
26	Mean characteristic impedance	(100 ± 5) ohm.
27	Nominal velocity of propagation	approx. 67%
28	Propagation delay	=<534 ns/100m
29	Delay skew	= < 40 ns/100m
30	Test voltage (DC, 1 min) Core/Core	1000 V.

Guaranteed Technical Particulars of CAT6 CABLE

1.	Manufacturers Name & Address:	
2.	Suppliers Name & Address:	
3.	Governing standards:	
4.	Type of cable:	
5	Category of cable:	
6	Conductor :	
7	No of conductor :	
8	Insulation: Polyethylene,	
9	Twisting:	
10	Cable lay up:	
11	Cu cable shielding :	
12	Stranding :	
13	Cable jacket material:	
14	Cable jacket charactertics:	
15	Cable overall diameter :	
16	Conductor diameter:	
17	Conductor Type:	
18	Length (meter):	
19	Tensile force :	
20	Mechanical properties Bending radius :	
21	Temperature range :	
22	Electrical properties DC loop resistance: (at 20°C ± 5°C): =<	
23	Resistance unbalance: =	
24	Insulation resistance (500 V) :	
25	Capacitance: 44 pF/m max	
26	Capacitance unbalance (pair to ground):	
27	Mean characteristic impedance:	
28	Nominal velocity of propagation:	
29	Propagation delay:	
30	Delay skew :	
31	Test voltage (DC, 1 min) Core/Core:	